

3 Humans store energy as glycogen.

(a) (i) Which is the correct statement about the formation of glycogen?

(1)

- ☐ **A** α glucose molecules join together by a condensation reaction
- ☐ **B** α glucose molecules join together by a hydrolysis reaction
- ☐ **C** β glucose molecules join together by a condensation reaction
- ☐ **D** β glucose molecules join together by a hydrolysis reaction

(ii) Name the bond that joins two glucose molecules together.

(1)

(iii) Explain how the structure of glycogen relates to its role as an energy storage molecule.

(3)

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(b) Von Gierke disease is one type of glycogen storage disease (GSD).

- (i) Between one in 20 000 and one in 25 000 babies are born with GSD.

About 25% of patients with GSD are thought to have Von Gierke disease.

In one country, 3.8 million babies were born in one year.

Estimate the number of babies born each year with Von Gierke disease in this country.

(2)

Answer

- (ii) Von Gierke disease is an inherited disease.

Suggest why the majority of these babies are born to parents who are not affected by Von Gierke disease.

(2)

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(Total for Question 3 = 9 marks)

